

# Carbon Calculator for Households

This project is developed by ,

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# Understanding Household Emissions

## The Problem

Households lack awareness of their CO<sub>2</sub> emissions. This absence of data hinders effective carbon reduction efforts. Tracking progress is challenging without precise measurements.

## Why It Matters

Climate change is an urgent global challenge. Collective small household changes create significant impact. This tool supports national and international carbon reduction goals.

## Our Objective

We aim to deliver a user-friendly carbon calculator. It measures household CO<sub>2</sub> emissions. It also recommends practical, sustainable actions. We focus on key emission areas.

# Our Innovative Solution



## Web-Based Calculator

Developed using JavaScript and D3.js. Users input data on electricity, travel, and diet. It calculates annual CO<sub>2</sub> emissions.



## Visual & Personalized

Provides clear visual output using charts. Offers personalized reduction tips. Examples: switching to LEDs or reducing meat intake.



## Innovation & Tools

Features a simple, interactive design. Encourages informed decision-making. Utilizes JavaScript, D3.js, HTML/CSS, and GitHub.



# SWOT Analysis & Insights

| Strengths                                                     | Weaknesses                                                        |
|---------------------------------------------------------------|-------------------------------------------------------------------|
| User-friendly, visual tool.<br>Promotes awareness and action. | Basic inputs only for now.<br>Assumes average emission factors.   |
| Opportunities                                                 | Threats                                                           |
| Expand to water/waste tracking.<br>Partner with schools/NGOs. | Competition from existing apps.<br>User reluctance to input data. |

Insights gained reveal that data visualization effectively drives behavior change. Personalizing tips significantly improves user trust and engagement with the tool. These insights will guide future development.

## SWOT





# Learnings & Next Steps

1

## Key Learnings

Applied JavaScript and D3.js in a real-world scenario. Understood how data influences sustainable actions. Gained experience in project planning, UI design, and testing.

2

## Future Enhancements

Add features for water and waste footprint. Implement a user login system for personalized tracking. Explore advanced data input options.

3

## Growth & Outreach

Promote the tool through schools and green clubs. Launch a mobile version for broader accessibility. Seek partnerships for wider adoption.

***Thank You***